Temperature and Concentration Compared with Benchmarks

> Part of the Functional Series. The platform's temperature series is a single dataset (Berkeley Earth, preliminary) and its concentration series is a single station. This document compares both with the IPCC, WMO, Copernicus, Berkeley Earth's own reports, the Forster et al. annual indicators and NOAA's global mean.

1. Temperature over multi-year periods

The IPCC assesses warming as multi-year averages, which is the right way to compare it with a single dataset. All values are relative to 1850–1900.

Period	Platform (Berkeley Earth HR)	Reference	Reference value (°C)
2001–2020	1.01	IPCC AR6 WGI SPM A.1.2	0.99 [0.84–1.10]
2010–2019	1.08	IPCC AR6 WGI SPM	1.06 [0.88–1.21]
2011–2020	1.11	IPCC AR6 WGI SPM A.1.2	1.09 [0.95–1.20]
2016–2025	1.29	Forster et al. 2026 (observed)	1.26 [1.13–1.36]
2023–2025	1.49	WMO consolidated, eight datasets	1.48 ± 0.13

The platform is within 0.02–0.03 °C of the IPCC and WMO values and well inside every reference range. Berkeley Earth is one of the eight datasets WMO consolidates, so the WMO row is agreement with a synthesis that includes it, not an independent confirmation.

2. Temperature in single years

Year	Platform	Berkeley Earth's own report	WMO consolidated (8 datasets)	Copernicus ERA5	Forster et al. (observed)
2023	1.48	about 1.47 (derived)			
2024	1.55	about 1.52 (derived)		1.60	1.51 ± 0.13
2025	1.45	1.44 ± 0.09	1.44 ± 0.13	1.47	1.39 ± 0.13

Notes:

- **The spread across datasets in a single year is about 0.1 °C** (2025 runs from 1.39 to 1.47), comparable to the stated uncertainties. The IPCC puts the internal variability of a single year at about ±0.25 °C (5–95% range), which is why it reports warming as running 20-year averages. A difference of a few hundredths of a degree between the platform and any one reference is not meaningful.

- **Berkeley Earth's own report** gives 2024 and 2023 only implicitly (2025 is "about 0.08 °C cooler" than 2024 and "0.03 °C cooler" than 2023), so those two entries are derived. The platform's 2024 value is about 0.03 °C above the derived figure; Berkeley Earth notes that its analyses are revised as new observations arrive, typically by small amounts.
- **2025 ranks third warmest** in six of the eight WMO datasets and second in two; the platform agrees on the ranking (2024 > 2023 > 2025 in its series).
- **The single warm years are not the long-term level.** 2023–2025 averaged above 1.5 °C in Copernicus's analysis, which presents it as the first three-year period to do so. The Paris temperature goal is about the long-term level, which is lower (§3).

3. Long-term warming, and when it reaches 1.5 °C

Several groups estimate the *underlying* warming level, free of year-to-year variability:

Source	Estimate	Method
Forster et al. 2026	Human-induced warming 1.37 °C in 2025 [1.1–1.7]; rising 0.27 °C a decade; reaches 1.5 °C around 2030	Attribution of observed warming to human causes
Copernicus	about 1.4 °C	"Several methods" (not named on the page read)
Berkeley Earth	Long-term trend "has already crossed 1.4 °C"; "less than 5 years" to reach 1.5 °C	30-year LOESS smooth
Global Carbon Budget 2025	Human-caused warming 1.36 °C in 2024	Attribution of observed warming to human activities
Platform	Starting level 1.32 °C (mean of 2020–2024); implied 2025 level 1.34 °C	Five-year mean of observations plus implied warming

The platform's starting level is lower than the others because a *five-year mean of observations* lags a trend that is still rising, and because it is an observed value, not an attributed one. The platform's implied 2025 level (1.34) is about 0.03 °C below Forster et al.'s human-induced 1.37, consistent with that.

4. Atmospheric CO_2

The platform uses the Mauna Loa station for its recent record. The IPCC, the Global Carbon Budget and NOAA's global series use a **global mean**, so the like-for-like comparison is with NOAA's global annual mean.

Year	IPCC / Global Carbon Budget	NOAA global mean	Platform (Mauna Loa)	Platform minus global
2019	410 ppm (IPCC SYR A.1.3)	410.07	411.65	+1.58
2024	422.8 ± 0.1 (GCB 2025)	422.79	424.61	+1.82
2025	425.6 (GCB 2025 projection)	425.62	427.35	+1.73

The platform series is about 1.6–1.8 ppm above the global mean every year, the expected offset of a Northern Hemisphere station. The global means agree with each other to within 0.05 ppm, and the platform's trend (about

2–3 ppm a year) matches. One further reference, quoted in the UNEP report from WMO, gives 423.9 ppm for 2024; WMO's global-mean product differs slightly from NOAA's.

5. What this comparison shows

- **Temperature: agreement within the datasets' own spread**, both over decades and in recent years. The platform's latest values are preliminary and could move by a few hundredths.
- **Concentration: agreement once station and global means are matched.** The offset is systematic, not an error.
- **Long-term warming is the figure that matters for Paris, and it is below the recent single years.** The platform's scenario translation starts at 1.32 °C for that reason.

See also

- The Reference Landscape
- Emissions Compared with Benchmarks
- Pathways, Carbon Budgets and Warming Outlooks
- Concentrations and Temperature: how the platform's two series are built